Mahler Measure of ± 1 Polynomials and the Pandrosion Fiber Constraints

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April 2026

Abstract

We analyze the Mahler measure of Littlewood polynomials (± 1 coefficients) through the Pandrosion framework. The fibers at $z = 0, 1, -1$ give arithmetic constraints: $|P(0)| = 1$ forces $\prod_{|\alpha|\le 1} |\alpha| = 1/M(P)$; $|P(1)| \ge 1$ and $|P(-1)| \ge 1$ constrain roots near ± 1 via the identities $\sum 1/((\alpha_j - z)P'(\alpha_j)) = -1/P(z)$. Combined with the vanishing $\sum 1/P'(\alpha_j) = 0$, these couple the roots inside and outside the unit circle. We verify numerically that *low Mahler measure forces* $|P(1)| = 1$ *or* $|P(-1)| = 1$, and that discriminants of ± 1 polynomials grow rapidly with degree.

1 Littlewood polynomials

Definition 1.1. A *Littlewood polynomial* of degree n is $P(z) = \sum_{k=0}^{n} \varepsilon_k z^k$ with $\varepsilon_k \in \{-1, +1\}$, $\varepsilon_n = 1$ (monic normalization).

The Mahler measure is $M(P) = \prod_j \max(1, |\alpha_j|)$, and the *Parseval bound* gives $M(P) \le \sqrt{n+1}$.

2 Pandrosion fiber constraints

Theorem 2.1 (Triple fiber constraint)**.** *For any Littlewood polynomial:*

$$\text{(Fiber at 0)} \quad \sum_{j=1}^{n} \frac{1}{\alpha_j \, P'(\alpha_j)} = -\frac{1}{P(0)} = \pm 1, \tag{1}$$

$$\text{(Fiber at 1)} \quad \sum_{j=1}^{n} \frac{1}{(\alpha_j - 1)\, P'(\alpha_j)} = -\frac{1}{P(1)}, \tag{2}$$

$$\text{(Fiber at } -1\text{)} \quad \sum_{j=1}^{n} \frac{1}{(\alpha_j + 1)\, P'(\alpha_j)} = -\frac{1}{P(-1)}. \tag{3}$$

Proposition 2.2. $P(1) = \sum_{k=0}^{n} \varepsilon_k$ *has the same parity as* $n+1$*, so* $|P(1)| \ge 1$*. Similarly* $|P(-1)| \ge 1$*.*

Corollary 2.3 (Root–Mahler coupling)**.** *Since* $|P(0)| = |\varepsilon_0| = 1$ *and* $M(P) = \prod_{|\alpha_j|>1} |\alpha_j|$*:*

$$\prod_{|\alpha_j|\le 1} |\alpha_j| = \frac{|P(0)|}{M(P)} = \frac{1}{M(P)}. \tag{4}$$

The Pandrosion vanishing $\sum 1/P'(\alpha_j) = 0$ *couples this to the roots outside the circle.*

3 Low Mahler forces extremal evaluations

Proposition 3.1 (Numerical observation)**.** *Among* 20,000 *random Littlewood polynomials of each degree* $n = 8, 10, 12$*: those with lowest Mahler measure always satisfy* $\min(|P(1)|, |P(-1)|) = 1$ *(the minimal allowed value).*

n	$M(P)$	$P(1)$	$P(-1)$
8	1.000	1	9
10	1.000	11	1
12	1.000	13	1
12	1.615	-3	1

Remark 3.2. The minimum $M(P) = 1$ is achieved by cyclotomic-type Littlewood polynomials (all roots on the unit circle). For non-cyclotomic Littlewood polynomials, the minimum Mahler measure appears to be ≈ 1.615 (golden ratio).

4 Discriminant growth

Proposition 4.1. *The discriminants of* ± 1 *polynomials are integers that grow rapidly:*

n	$\min \lvert\text{disc}\rvert$	$\max \lvert\text{disc}\rvert$
4	*117*	*605*
6	*713*	*927 369*
8	*59 049*	2.4×10^9

This means $\prod |P'(\alpha_j)| \geq \sqrt{117} \approx 10.8$ already for $n = 4$, giving a strong constraint on root separation.

5 Root distribution

Proposition 5.1. *For random Littlewood polynomials: the number of roots inside the unit circle is approximately* $n/2$ *on average.*

n	*Avg. roots inside* $\lvert z\rvert < 1$
6	*3.00 (50.0%)*
10	*4.99 (49.9%)*
15	*7.46 (49.7%)*

By Corollary 2.3, $\prod_{|\alpha| \leq 1} |\alpha| = 1/M(P)$*, so the* $\approx n/2$ *inside roots conspire to produce* $M(P) \geq 1$*.*

6 The Pandrosion picture

Point	**Fiber identity**	± 1 **constraint**
$z = 0$	$\sum 1/(\alpha_j P') = \pm 1$	$\lvert P(0)\rvert = 1$ always
$z = 1$	$\sum 1/((\alpha - 1)P') = -1/P(1)$	$\lvert P(1)\rvert \geq 1$, odd
$z = -1$	$\sum 1/((\alpha + 1)P') = -1/P(-1)$	$\lvert P(-1)\rvert \geq 1$, odd
$z = e^{i\theta}$	general fiber	Parseval: $\int \lvert P\rvert^2 = n + 1$

The Pandrosion fibers at the three distinguished points $0, 1, -1$ provide arithmetic constraints that, combined with the vanishing identity and Parseval, severely restrict the Mahler measure of Littlewood polynomials.

References

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